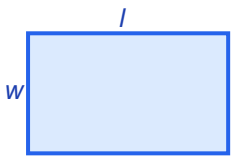
RAG Learning

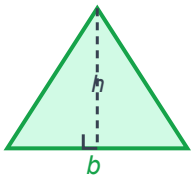
4.2 Mensuration: Area, Perimeter and Volume

OCR GCSE Mathematics J560 | Component 4 (Geometry and Measures) | Papers 1, 2 & 3

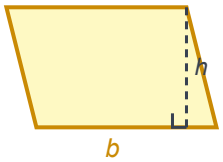
4.2.1 Perimeter and Area of Standard Shapes



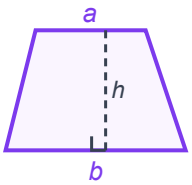
Rectangle



Triangle



Parallelogram



Trapezium

Shape	Area Formula	Perimeter
Rectangle	$A = lw$	$P = 2(l + w)$
Triangle	$A = \frac{1}{2}bh$	Sum of three sides
Parallelogram	$A = bh$ (perpendicular height)	$P = 2(a + b)$
Trapezium	$A = \frac{1}{2}(a + b)h$	Sum of four sides

Exam Tip: For the triangle and parallelogram, h must be the **perpendicular** height — the vertical distance between the parallel sides, not the slant side.

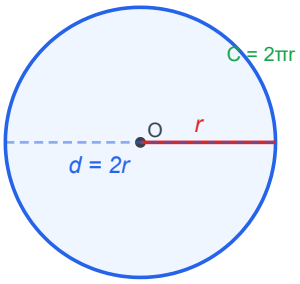
Worked Example: Find the area of a trapezium with parallel sides 8 cm and 14 cm and perpendicular height 5 cm.

$A = \frac{1}{2}(8 + 14) \times 5 = \frac{1}{2} \times 22 \times 5 = \mathbf{55}\,\mathrm{cm}^{\mathbf{2}}$

Worked Example: A triangle has base 9 cm and area 27 cm². Find the perpendicular height.

$27 = \frac{1}{2} \times 9 \times h \Rightarrow h = 27 \times 2 \div 9 = \mathbf{6}\,\mathrm{cm}$

4.2.2 Area and Circumference of Circles



$$\text{Circumference: } C = 2\pi r = \pi d$$

$$\text{Area: } A = \pi r^2$$

For an arc of angle θ°: $\text{Arc length} = \dfrac{\theta}{360} \times 2\pi r$

For a sector of angle θ°: $\text{Sector area} = \dfrac{\theta}{360} \times \pi r^2$

Worked Example: Find the circumference and area of a circle with radius 7 cm. (Leave answers in terms of π.)

$C = 2\pi \times 7 = 14\pi \approx \mathbf{44.0}\,\mathrm{cm}$
$A = \pi \times 7^2 = 49\pi \approx \mathbf{153.9}\,\mathrm{cm}^{\mathbf{2}}$

Worked Example: A sector has radius 12 cm and angle 150°. Find its arc length and area.

Arc length = $\frac{150}{360} \times 2\pi \times 12 = \frac{5}{12} \times 24\pi = 10\pi \approx \mathbf{31.4}\,\mathrm{cm}$

Sector area = $\frac{150}{360} \times \pi \times 12^2 = \frac{5}{12} \times 144\pi = 60\pi \approx \mathbf{188.5}\,\mathrm{cm}^{\mathbf{2}}$

Exam Tip: If the question says "leave in terms of π", do not press the π button on your calculator — just write 49π etc. If it asks for a decimal, use $\pi \approx 3.14159\ldots$

4.2.3 Area of Composite Shapes

A **composite shape** is made up of two or more standard shapes joined together. Split it into rectangles, triangles, circles, etc. and add or subtract areas.

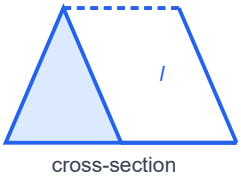
Worked Example: Find the area of an L-shaped figure formed by a rectangle 10 cm × 8 cm with a 4 cm × 3 cm rectangle cut from one corner.

Area of full rectangle: $10 \times 8 = 80\,\mathrm{cm}^2$
Area removed: $4 \times 3 = 12\,\mathrm{cm}^2$
Composite area = $80 - 12 = \mathbf{68}\,\mathrm{cm}^{\mathbf{2}}$

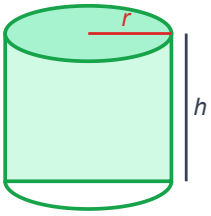
Worked Example: A shape consists of a rectangle 6 cm × 4 cm with a semicircle of diameter 4 cm attached to one end. Find the total area.

Rectangle area = $6 \times 4 = 24\,\mathrm{cm}^2$
Semicircle: radius = 2 cm; area = $\frac{1}{2}\pi \times 2^2 = 2\pi \approx 6.28\,\mathrm{cm}^2$
Total = $24 + 2\pi \approx \mathbf{30.3}\,\mathrm{cm}^{\mathbf{2}}$

4.2.4 Volume and Surface Area of Prisms and Cylinders



Triangular prism (length l)



Cylinder

All prisms and cylinders:

$$V = A_{\text{cross-section}} \times l$$

Shape	Volume	Curved Surface Area	Total Surface Area
Prism (general)	$V = A \times l$	Perimeter of cross-section $\times l$	$2A + \text{CSA}$
Cylinder	$V = \pi r^2 h$	$2\pi rh$	$2\pi rh + 2\pi r^2$

Worked Example: Find the volume and total surface area of a cylinder with radius 5 cm and height 12 cm.

Volume: $V = \pi \times 5^2 \times 12 = 300\pi \approx \mathbf{942.5}\,\text{cm}^{\mathbf{3}}$

Curved surface area: $2\pi \times 5 \times 12 = 120\pi \approx 376.99\,\text{cm}^2$
Two circular ends: $2 \times \pi \times 5^2 = 50\pi \approx 157.08\,\text{cm}^2$
Total SA: $120\pi + 50\pi = 170\pi \approx \mathbf{534.1}\,\text{cm}^{\mathbf{2}}$

Worked Example: A triangular prism has a right-angled triangular cross-section with legs 3 cm and 4 cm (hypotenuse 5 cm). The prism has length 10 cm. Find its volume and surface area.

Volume: Cross-section area = $\frac{1}{2} \times 3 \times 4 = 6\,\text{cm}^2$

$V = 6 \times 10 = \mathbf{60}\,\mathrm{cm}^{\mathbf{3}}$

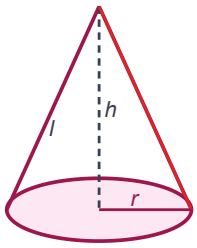
Surface area:

2 triangular ends: $2 \times 6 = 12\,\mathrm{cm}^2$

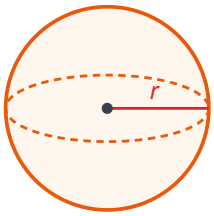
3 rectangular faces: $(3 \times 10) + (4 \times 10) + (5 \times 10) = 30 + 40 + 50 = 120\,\mathrm{cm}^2$

Total SA = $12 + 120 = \mathbf{132}\,\mathrm{cm}^{\mathbf{2}}$

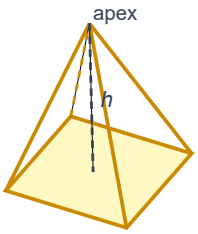
4.2.5 Volume and Surface Area of Pyramids, Cones and Spheres



Cone



Sphere



Square-based pyramid

Shape	Volume	Surface Area
Pyramid (any base)	$V = \frac{1}{3}Ah$	Base area + area of all triangular faces
Cone	$V = \frac{1}{3}\pi r^2 h$	$\pi r l + \pi r^2$ where $l = \sqrt{r^2 + h^2}$
Sphere	$V = \frac{4}{3}\pi r^3$	$4\pi r^2$

l = slant height of cone; h = perpendicular height; A = base area of pyramid.

Worked Example: Find the volume of a cone with radius 6 cm and height 8 cm.

$$V = \frac{1}{3} \times \pi \times 6^2 \times 8 = \frac{1}{3} \times 36\pi \times 8 = 96\pi \approx \mathbf{301.6}\,\mathrm{cm}^{\mathbf{3}}$$

Worked Example: Find the total surface area of a cone with radius 5 cm and height 12 cm.

Slant height: $l = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13\,\text{cm}$
Curved SA = $\pi \times 5 \times 13 = 65\pi$
Base area = $\pi \times 5^2 = 25\pi$
Total SA = $65\pi + 25\pi = 90\pi \approx \mathbf{282.7}\,\text{cm}^{\mathbf{2}}$

Worked Example: A sphere has volume $288\pi\ \text{cm}^3$. Find its radius.

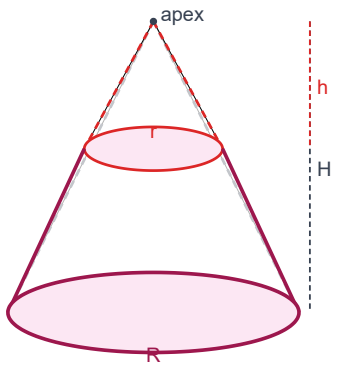
$$\frac{4}{3}\pi r^3 = 288\pi$$
$$r^3 = 288 \times \frac{3}{4} = 216$$
$$r = \sqrt[3]{216} = \mathbf{6}\,\text{cm}$$

Worked Example: A square-based pyramid has base side 9 cm and perpendicular height 10 cm. Find its volume.

Base area $A = 9^2 = 81\,\text{cm}^2$
$V = \frac{1}{3} \times 81 \times 10 = \mathbf{270}\,\text{cm}^{\mathbf{3}}$

4.2.6 Frustums Higher

A **frustum** is the part of a cone left after the top (a smaller, similar cone) is removed by a cut parallel to the base.



Volume of frustum = Volume of large cone − Volume of small cone:

$$V_{\text{frustum}} = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi r^2 h = \frac{\pi}{3}(R^2 H - r^2 h)$$

where R = radius of base, r = radius of top, H = height of full cone, h = height of removed small cone.

Exam Tip: To find H and h, use the similar triangle ratio $\frac{r}{R} = \frac{h}{H}$. If you are given only the height of the frustum ($H - h$), you will need to set up a ratio to find H and h separately.

Worked Example: A frustum is formed by removing a small cone of height 6 cm and radius 2 cm from a full cone of height 18 cm and radius 6 cm. Find the volume of the frustum.

Check similarity ratio: $r/R = 2/6 = 1/3$ and $h/H = 6/18 = 1/3$ ✓ (consistent)

$V_{\text{large}} = \frac{1}{3}\pi \times 6^2 \times 18 = \frac{1}{3}\pi \times 36 \times 18 = 216\pi$
$V_{\text{small}} = \frac{1}{3}\pi \times 2^2 \times 6 = \frac{1}{3}\pi \times 4 \times 6 = 8\pi$
$V_{\text{frustum}} = 216\pi - 8\pi = 208\pi \approx \mathbf{653.5}\,\text{cm}^3$

4.2.7 Units of Area and Volume

Units of length: $1\,\text{km} = 1000\,\text{m}$; $1\,\text{m} = 100\,\text{cm}$; $1\,\text{cm} = 10\,\text{mm}$

Units of area:

$$1\,\mathrm{m}^2 = 10{,}000\,\mathrm{cm}^2 = 100^2\,\mathrm{cm}^2$$

$$1\,\mathrm{km}^2 = 1{,}000{,}000\,\mathrm{m}^2$$

$$1\,\text{hectare (ha)} = 10{,}000\,\mathrm{m}^2$$

Units of volume:

$$1\,\mathrm{m}^3 = 1{,}000{,}000\,\mathrm{cm}^3 = 100^3\,\mathrm{cm}^3$$

$$1\,\text{litre} = 1000\,\mathrm{cm}^3 = 1000\,\mathrm{ml}$$

$$1\,\mathrm{m}^3 = 1000\,\text{litres}$$

Conversion	Multiply by	Example
cm^2 to mm^2	$\times 100$	$5\,\mathrm{cm}^2 = 500\,\mathrm{mm}^2$
m^2 to cm^2	$\times 10{,}000$	$3\,\mathrm{m}^2 = 30{,}000\,\mathrm{cm}^2$
cm^3 to mm^3	$\times 1000$	$2\,\mathrm{cm}^3 = 2000\,\mathrm{mm}^3$
m^3 to cm^3	$\times 1{,}000{,}000$	$0.5\,\mathrm{m}^3 = 500{,}000\,\mathrm{cm}^3$
cm^3 to litres	$\div 1000$	$5000\,\mathrm{cm}^3 = 5\,\text{litres}$
litres to cm^3	$\times 1000$	$2.5\,\text{litres} = 2500\,\mathrm{cm}^3$

Exam Tip: Remember: when converting units of *area*, square the linear conversion factor. When converting units of *volume*, cube the linear conversion factor. E.g., $1\,\mathrm{m} = 100\,\mathrm{cm}$, so $1\,\mathrm{m}^2 = 100^2 = 10{,}000\,\mathrm{cm}^2$ and $1\,\mathrm{m}^3 = 100^3 = 1{,}000{,}000\,\mathrm{cm}^3$.

Worked Example: A tank has volume $0.36\,\mathrm{m}^3$. How many litres of water does it hold?

$0.36\,\mathrm{m}^3 = 0.36 \times 1{,}000{,}000\,\mathrm{cm}^3 = 360{,}000\,\mathrm{cm}^3$
Litres = $360{,}000 \div 1000 = \mathbf{360}\,\text{litres}$

Master Formula Summary

Shape	Area / Volume	Perimeter / Surface Area
Rectangle	$A = lw$	$P = 2(l + w)$
Triangle	$A = \frac{1}{2}bh$	Sum of sides
Parallelogram	$A = bh$	$2(a + b)$
Trapezium	$A = \frac{1}{2}(a + b)h$	Sum of sides
Circle	$A = \pi r^2$	$C = 2\pi r$
Sector	$A = \frac{\theta}{360}\pi r^2$	$\frac{\theta}{360} \times 2\pi r + 2r$
Prism	$V = A_{\text{cs}} \times l$	$2A_{\text{cs}} + P_{\text{cs}} \times l$
Cylinder	$V = \pi r^2 h$	$2\pi rh + 2\pi r^2$
Pyramid	$V = \frac{1}{3}Ah$	Base + triangular faces
Cone	$V = \frac{1}{3}\pi r^2 h$	$\pi rl + \pi r^2; l = \sqrt{r^2 + h^2}$
Sphere	$V = \frac{4}{3}\pi r^3$	$SA = 4\pi r^2$
Frustum (Higher)	$V = \frac{\pi}{3}(R^2 H - r^2 h)$	$\pi Rl + \pi rl + \pi R^2 + \pi r^2$ (approx.)